

Carlos Andrés Canamejoy Alegría

Generative AI | LLMs & Agentic AI | Deep Learning | Computer Vision
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PROFESSIONAL PROFILE

Physics Engineer with 2+ years of research experience in deep learning, generative AI, and computer vision. Skilled in designing AI agent workflows, LLM-based solutions, and intelligent automation using Python, PyTorch, and TensorFlow. Hands-on experience with Claude, Claude Code, prompt engineering, and tool use / function calling patterns. Proven ability to build end-to-end prototypes and proofs of concept, from data preparation through model training to evaluation. Practical knowledge of cloud platforms (AWS, Databricks), MLOps pipelines (Airflow, Docker, CI/CD), and data engineering tools (Spark, Hadoop, Snowflake). Passionate about translating emerging AI capabilities into practical solutions that create measurable business value. Strong collaborative mindset with experience in multidisciplinary research teams.

SKILLS

- Analytical reasoning and structured problem solving.
- Rapid prototyping, experimentation, and continuous learning.
- Cross-functional collaboration in multidisciplinary teams.
- Technical communication, documentation, and knowledge sharing.
- Time management and task prioritization.

TECHNICAL SKILLS

- **Programming:** Python, SQL.
- **Generative AI & LLMs:** LLMs, Agentic AI, AI Agents, Prompt Engineering, Tool Use / Function Calling, Conversational Memory, Intelligent Orchestration.
- **AI Platforms & Tools:** Claude, Claude Code, Claude Skills, Custom GPT Assistants.
- **Machine Learning / Deep Learning:** PyTorch, TensorFlow, scikit-learn.
- **Computer Vision:** CNNs, image processing, generative models (GANs, VAEs, Diffusion Models).
- **MLOps & DevOps:** Training/validation pipelines, CI/CD for ML, model deployment, Airflow.
- **Cloud & Data Platforms:** AWS (EC2, S3), Databricks.
- **Containers & Orchestration:** Docker, Kubernetes.
- **Data Engineering:** Spark, Hadoop, Snowflake.
- **Data Analysis:** NumPy, Pandas, Matplotlib.
- **Infrastructure:** Linux, Nginx.
- **Version Control:** Git, GitHub.
- **Development Tools:** Jupyter Notebook, Google Colab, L^AT_EX.
- **UI/UX:** Figma.

EXPERIENCE

Research Assistant

March 2026 – June 2026

Signal Control and Digital Signal Processing Research Group (A1 Category, Minciencias) – National University of Colombia

- Designed and developed end-to-end generative AI pipelines (conditional VAEs, diffusion models) in PyTorch and TensorFlow, building prototypes and PoCs for scientific image synthesis from physical parameters.
- Trained, evaluated, and fine-tuned convolutional neural networks (CNNs) to estimate physical parameters from microscopy images, validating predictions against ground-truth simulations.
- Built AI agent workflows leveraging LLMs, prompt engineering, and tool use / function calling patterns for research task automation.
- Developed reproducible MLOps pipelines (training, validation, deployment) with Airflow orchestration, reducing experiment iteration time.
- Integrated cloud services (AWS EC2, S3, Databricks) into AI workflows for scalable data processing and model training.
- Curated and validated image datasets, ensuring data quality for model training and evaluation.
- Documented architectures, implementation patterns, and learnings for the research team.
- Co-authored a scientific article on physics-informed generative models (currently under preparation).

Laboratory Assistant Fellow – Physics Laboratories

February 2025 – December 2025

National University of Colombia, Manizales Campus

- Facilitated experimental sessions in Mechanics, Electricity and Magnetism, and Waves and Optics for undergraduate students.
- Mentored students in experimental design, data collection, and quantitative result analysis.
- Coordinated maintenance and inventory tracking of laboratory equipment and instrumentation.

Technology Intern

March 2020 – May 2020

Municipality of Puerto Caicedo, Putumayo

- Delivered technical support for hardware and software across the organization.
- Configured and deployed local area networks for internal operations.
- Led training sessions on productivity and technology tools for non-technical staff.

EDUCATION

M.Sc. in Engineering – Industrial Automation

Expected (2027)

Universidad Nacional de Colombia

Physics Engineering

2026

Universidad Nacional de Colombia

Information Systems Technician

2020

Servicio Nacional de Aprendizaje (SENA)

LANGUAGES

- **Spanish:** Native.
- **English:** B1.

COURSES AND CERTIFICATIONS

- **Microsoft AI Skills Fest 2026** – Microsoft. Badge.
- **AI Agents and Agentic AI with Python & Generative AI** – Coursera.
- **Claude Code: Software Engineering with Generative AI Agents** – Coursera.
- **Introduction to Generative AI** – Coursera.
- **Prompt Engineering for ChatGPT** – Coursera.
- **Spark, Hadoop, and Snowflake for Data Engineering** – Coursera.
- **Virtualization, Docker, and Kubernetes for Data Engineering** – Coursera.
- **Python Essentials for MLOps** – Coursera.
- **Access an EC2 instance shell from the AWS console** – Coursera.
- **AWS S3 Basics** – Coursera.
- **Data Visualization with Python** – Coursera.

REFERENCES

Andrés Marino Álvarez Meza

Co-Director, Signal Control and Digital Signal Processing Research Group – Senior Researcher, Minciencias

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Graph Data Science – PhD Cand.

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